Topic 3 – Data representation

Students should:		Curriculum reference
3.1	Be able to convert between binary and denary positive integers (0–255)	DR7.1D
3.2	Be able to add two positive binary numbers	DR8.1A
3.3	Understand how computers use binary to represent data (numbers, text, sound, graphics) and program instructions	DR7.1A DR8.1B, C, D DR9.1A
3.4	Understand why and how computers encode characters using ASCII and Unicode	DR7.1E
3.5	Understand how bitmap images are represented in binary (pixels, resolution, colour depth (2-bit max))	DR8.1B, C, D
3.6	Understand how sound is represented in binary (sample rate, amplitude, bit-depth)	DR9.1A, B
3.7	Understand the terms describing capacity of storage (bit, byte, kibibyte, mebibyte, gibibyte)	DR8.1E, G
3.8	Be able to convert storage capacities into different units of measurement	DR8.1F, G